

Experiment 12: Iris Flower Classification using KNN**Aim**

To implement **Iris Flower Classification using the K-Nearest Neighbours (KNN) algorithm** in Python.

Algorithm

Step 1: Import the required libraries.

Step 2: Load the Iris dataset.

Step 3: Separate the input features and target classes.

Step 4: Split the dataset into training and testing data.

Step 5: Create a KNN classifier with a suitable value of K.

Step 6: Train the model using the training data.

Step 7: Predict the flower classes for the test data.

Step 8: Calculate the classification accuracy.

Step 9: Display the predictions and accuracy.

Program

```
# Experiment 12 - Iris Flower Classification using KNN

from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score

iris = load_iris()

X = iris.data
y = iris.target

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

model = KNeighborsClassifier(n_neighbors=3)
model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print("Predictions:", y_pred)
print("Accuracy:", accuracy_score(y_test, y_pred))
```

Output

```
... Predictions: [1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 1 2 1 1 2 0 2 0 2 2 2 2 2 0 0]  
Accuracy: 1.0
```

Result

Thus, the **Iris Flower Classification using KNN was successfully implemented**, and the Iris flower classes were predicted using the trained KNN model.